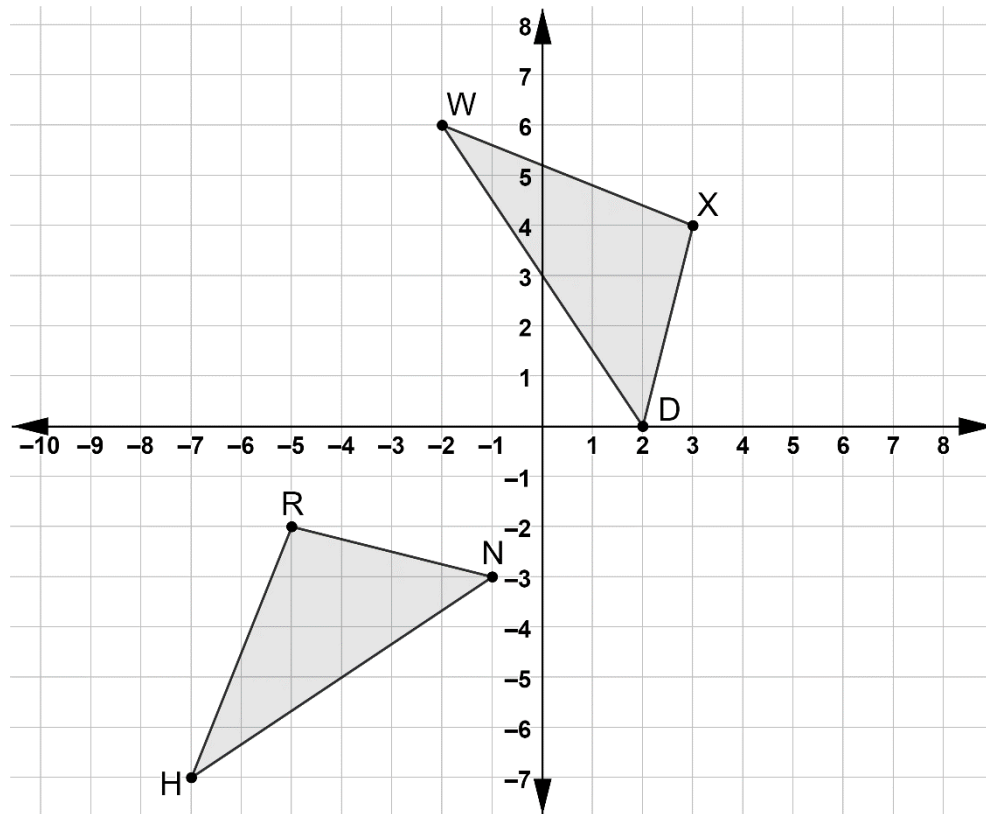


Triangles WXD and RNH are shown. Assume $\overline{WX} \cong \overline{HR}$, $\overline{RN} \cong \overline{XD}$, and $\overline{DW} \cong \overline{NH}$.



1. Explain how to use geometric tools to verify that $\triangle HRN \cong \triangle WXD$.
2. Write a sequence of transformations that will map $\triangle HRN$ onto $\triangle WXD$.

Show your work for questions 3 through 5 on the coordinate grid.

3. Draw $\triangle H'R'N'$ the result of the first transformation.
4. Map the vertices of $\triangle HRN$ to $\triangle H'R'N'$.
5. Map the vertices of $\triangle H'R'N'$ to $\triangle WXD$.

For questions 6 through 8, complete the statements.

6. The outcome of mapping the transformations of $\triangle HRN$ to $\triangle H'R'N'$ and then from $\triangle H'R'N'$ to $\triangle$ _____ shows that point H' coincides with point _____, point R' coincides with point _____, and point N' coincides with point _____.

7. Because the definition of _____ in terms of _____
_____ preserves _____, it can be stated that
 $\triangle HRN \cong \triangle$ _____.

8. Since it was assumed that $\overline{WX} \cong \overline{HR}$, $\overline{RN} \cong \overline{XD}$, and $\overline{DW} \cong \overline{NH}$ then _____
is sufficient to show triangles are congruent.

9. Select all of the given statements that would result in a sequence of transformations that would show SAS congruence is sufficient to show the triangles are congruent.

- ☐ $\overline{WX} \cong \overline{HR}$, $\overline{RN} \cong \overline{XD}$, $\angle RNH \cong \angle XDW$
- ☐ $\overline{RN} \cong \overline{XD}$, $\overline{DW} \cong \overline{NH}$, $\angle RNH \cong \angle XDW$
- ☐ $\overline{WX} \cong \overline{HR}$, $\overline{DW} \cong \overline{NH}$, $\angle RHN \cong \angle XWD$
- ☐ $\overline{WX} \cong \overline{HR}$, $\overline{RN} \cong \overline{XD}$, $\angle NHR \cong \angle XWD$
- ☐ $\overline{RN} \cong \overline{XD}$, $\overline{DW} \cong \overline{NH}$, $\angle NHR \cong \angle XWD$
- ☐ $\overline{WX} \cong \overline{HR}$, $\overline{DW} \cong \overline{NH}$, $\angle NHR \cong \angle XWD$

10. Select all of the given statements that would result in a sequence of transformations that would show ASA congruence is sufficient to show the triangles are congruent.

- ☐ $\overline{WX} \cong \overline{HR}$, $\angle RNH \cong \angle XDW$, $\angle RHN \cong \angle XWD$
- ☐ $\overline{DW} \cong \overline{NH}$, $\angle RNH \cong \angle XDW$, $\angle NHR \cong \angle XWD$
- ☐ $\overline{WX} \cong \overline{HR}$, $\angle RHN \cong \angle XWD$, $\angle NRH \cong \angle DXW$
- ☐ $\overline{RN} \cong \overline{XD}$, $\angle RNH \cong \angle XDW$, $\angle NRH \cong \angle DXW$
- ☐ $\overline{DW} \cong \overline{NH}$, $\angle NHR \cong \angle XWD$, $\angle NRH \cong \angle DXW$
- ☐ $\overline{RN} \cong \overline{XD}$, $\angle NHR \cong \angle XWD$, $\angle NRH \cong \angle DXW$